

CLINICAL GUIDANCE



Multi-disciplinary collaborative consensus guidance statement on the assessment and treatment of mental health symptoms in patients with post-acute sequelae of SARS-CoV-2 infection (PASC)

Abby L. Cheng MD, MPHS¹ | Jordan Anderson DO² | Nyaz Didehbani PhD³ | Jeffrey S. Fine MD⁴ | Talya K. Fleming MD⁵ | Rasika Karnik MD⁶ | Michele Longo MD, MPH⁷ | Rowena Ng PhD^{8,9} | Yochai Re'em MD¹⁰ | Sarah Sampsel MPH¹¹ | Julieanne Shulman PsyD¹² | Julie K. Silver MD¹³ | Jamie Twaite PhD¹² | Monica Verduzco-Gutierrez MD¹⁴ | Monica Kurylo PhD, ABPP¹⁵

¹Division of Physical Medicine and Rehabilitation, Washington University, St. Louis, Missouri, USA

²School of Medicine, OHSU, Portland, Oregon, USA

³Departments of Psychiatry and Physical Medicine & Rehabilitation at UT Southwestern Medical Center, Dallas, Texas, USA

⁴Department of Rehabilitation Medicine, Rusk Rehabilitation, NYU Langone Health, New York, New York, USA

⁵Department of Physical Medicine and Rehabilitation, JFK Johnson Rehabilitation Institute at Hackensack Meridian Health, Edison, New Jersey, USA

⁶Department of Medicine, University of Chicago, Chicago, Illinois, USA

⁷Department of Clinical Neurosciences, Tulane University, New Orleans, Louisiana, USA

⁸Neuropsychology Department, Kennedy Krieger Institute, Johns Hopkins University School of Medicine, Baltimore, Maryland, USA

⁹Department of Psychiatry and Behavioral Sciences, Johns Hopkins University School of Medicine, Baltimore, Maryland, USA

¹⁰Department of Psychiatry, Weill Cornell Medicine, New York, New York, USA

¹¹SLSampsel Consulting, LLC, Albuquerque, New Mexico, USA

¹²The Arthur S. Abramson Department of Physical Medicine and Rehabilitation, Albert Einstein College of Medicine, Bronx, New York, USA

¹³Department of Physical Medicine and Rehabilitation, Harvard Medical School, Spaulding Rehabilitation Hospital, Massachusetts General Hospital, and Brigham and Women's Hospital, Boston, Massachusetts, USA

¹⁴Department of Rehabilitation Medicine, University of Texas Health Science Center at San Antonio, San Antonio, Texas, USA

¹⁵Neurorehabilitation Psychology Services, University of Kansas Medical Center (KUMC) & Kansas University Health System, Kansas City, Kansas, USA

Correspondence

Sarah Sampsel, SLSampsel Consulting, LLC, Albuquerque, NM, USA.

Email: slsampsel@ssampselqly.com

INTRODUCTION

Post-acute sequelae of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection (PASC), also called Long COVID (coronavirus disease), is the experience of new or worsened signs, symptoms, or conditions that develop after resolution of the acute phase of a COVID-19 infection. Although some people with COVID-19 recover well, others have persisting symptoms.¹ PASC can manifest as a wide-ranging

constellation of disabling sequelae, including mental health conditions.² Anxiety and depression have been reported as the second and third most common symptoms of PASC, respectively.³ Furthermore, in a narrative review summarizing neuropsychiatric dimensions of PASC, anxiety, post-traumatic stress disorder (PTSD), and depression were among the most reported symptoms of both ongoing symptomatic COVID-19 and PASC. Pooled prevalence of each was: anxiety 19.1% (95% confidence interval [CI], 13.3%–26.8%),

PTSD 15.7% (95% CI, 9.9%–24.1%), and depression 12.9% (95% CI, 7.5%–21.5%).⁴ Despite the high prevalence and often disabling impact of PASC and PASC-related mental health symptoms and the emerging data that PASC can persist for months or years,⁵ limited guidance currently exists regarding the assessment and treatment of mental health conditions in patients with PASC.

Addressing mental health symptoms in the setting of PASC involves several unique complexities, including addressing stigma that may interfere with appropriate diagnosis(es) and treatment, insufficient availability of mental health professionals, and differentiating mental and physical health diagnoses. Many patients have described being questioned about their PASC-related symptoms in a way that feels dismissive of their experience and/or mistakenly attributed to an underlying mental health condition.⁶ Although the effects of the COVID-19 pandemic on individuals and society can exacerbate direct sequelae from SARS-CoV-2 infection, increasingly evidence supports that new mental health symptoms can also be a component of PASC or caused by SARS-CoV-2 infection, and pre-existing mental health conditions can be exacerbated by PASC, yet mental health conditions are not, in and of themselves, the overall cause of PASC.^{7–9} Another challenge in addressing mental health disorders is the national and global shortage of mental health professionals.¹⁰ This challenge is further intensified because clinicians who are *not* mental health specialists often express discomfort and a perception of insufficient training to discuss mental health with their patients.

Nevertheless, most patients report that they want clinicians who are not mental health specialists to broach the topic and acknowledge the interplay between mental and physical health.^{11–13} Finally, studies have found that PASC can manifest as symptoms that are not due to a mental health condition, but that can mimic and/or be exacerbated by a mental health disorder. These symptoms include fatigue, dysautonomia, disordered sleep, and cognitive dysfunction,^{14,15} which can also interfere with a patient's ability to fully participate in first-line treatment recommendations.

The goal of this consensus statement is to present practical guidance for clinicians who treat patients with PASC. Specifically, this statement addresses the assessment and initial treatment of PASC-related mental health symptoms including depression, anxiety disorders (including panic), and PTSD. People with PASC have also reported new or worsening suicidal ideation, psychosis, obsessive compulsive disorder (OCD), and pandemic-related grief and survivor's remorse^{8,16–18}; however, specific focus on these conditions is outside the scope of this statement. The recommendations in this statement are applicable to *all* patients with PASC who are experiencing mental health symptoms, regardless of the time course. Of note, this guidance

statement reflects the current evidence base and related recommendations from an expert panel of health care professionals who regularly care for people with PASC. The recommendations should not preclude clinical judgment and must be applied in the context of each specific patient, with adjustments for patient preferences, comorbidities, and other factors.

PASC consensus guidance statement methods

The American Academy of Physical Medicine and Rehabilitation (AAPM&R) Multi-Disciplinary PASC Collaborative (PASC Collaborative) was convened to address the pressing need for guidance in the care of patients with PASC. PASC Collaborative members include experts across a variety of clinical disciplines and specialties including PM&R, neurology, neuropsychiatry, neuropsychology, rehabilitation psychology, and primary care. The PASC Collaborative is following an iterative, modified Delphi process to achieve consensus on assessment and treatment recommendations that have been presented as a series of consensus guidance statements regarding the most common PASC symptoms. A full description of the modified Delphi methodology has been published previously.¹⁹ These recommendations are informed by experts from established PASC centers who have experience managing patients with the full range of PASC-related symptoms.^{20–26} Whenever possible, existing evidence related to mental health assessment and treatment specifically for people with PASC has been incorporated into this guidance statement, and this body of evidence is somewhat limited and evolving. As needed, recommendations have also been guided by applying general, yet well-established, mental health assessment and treatment approaches to unique considerations in people with PASC. The consensus guidance statements include an intentional focus on health equity because disparities in care are a critical factor in contributing to widespread disparities in clinical outcomes, and they must be considered and addressed. The PASC Collaborative's goal is to broaden the understanding of current patient care practices, not only to guide clinical management, but also to identify critical gaps and opportunities for future research.

According to the original definition by the U.S. Centers for Disease Control and Prevention (CDC), “Long COVID (PASC) is broadly defined as signs, symptoms, and conditions that continue or develop after initial COVID-19 or SARS-CoV-2 infection. The signs, symptoms, and conditions are present 4 weeks or more after the initial phase of infection; may be multisystemic; and may present with a relapsing–remitting pattern and progression or worsening over time, with the possibility of severe and life-threatening events even months or

years after infection. The CDC uses the 4-week time-frame in describing post-COVID conditions to emphasize the importance of initial clinical evaluation and supportive care during the initial 4 to 12 weeks after acute COVID-19.²⁷ Based on patient feedback during our consensus process, we also advocate for improving access to beneficial interventions by facilitating early evaluation, diagnosis, and management of new or worsened symptoms that are experienced after COVID-19 infection. For the purposes of this guidance statement, we recommend assessment for PASC if symptoms are not improving within 1 month after acute symptom onset.

MENTAL HEALTH IN COVID-19 AND PASC

Terminology

Emotional and mood fluctuations are experienced as part of normal everyday life, but persistent, distressing disturbances in emotional regulation, behavior, and/or cognition can be indicative of a mental health disorder. According to the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) depressive disorders are characterized primarily by persistent sadness, emptiness, irritability, and/or a loss of pleasure or interest in activities (anhedonia). They are often associated with other secondary symptoms including poor concentration, psychomotor agitation or retardation, fatigue or loss of energy, sleep difficulty (insomnia or hypersomnia), feelings of low self-worth or excessive guilt, hopelessness about the future, changes in appetite or weight, and/or thoughts about dying or suicide. Anxiety disorders are characterized by excessive fear, worry, apprehension, or dread that is out of proportion to the perceived threat. PTSD is characterized by intrusive distressing memories, dreams, dissociative reactions (flashbacks), prolonged psychological distress, and/or physiological reactions that occur after exposure to an actual or threatened traumatic event.²⁸

Impact

The COVID-19 pandemic resulted in a 28% increase in cases of major depressive disorder (MDD) and a 26% increase in cases of anxiety disorders worldwide in 2020.^{29,30} It was also estimated to have caused 137 additional disability-adjusted life years (DALYs) per 100,000 people due to MDD and 116 additional DALYs per 100,000 people due to anxiety disorders, and the incidence of new mental health symptoms experienced after SARS-CoV-2 infection has been higher than after influenza or other respiratory infections.^{15,31,32} Both women and people from underrepresented race and ethnic minority groups have been reported to have a

higher incidence of PASC, regardless of the type(s) of PASC sequelae.^{33,34} Women also have a higher prevalence of pre-existing anxiety and autoimmune diseases than men, and both conditions are risk factors for developing PASC³⁵ and new or worsening mental health conditions. People who are at increased risk for specifically developing PASC-related mental health disorders include women, older adults, and potentially Black/African American individuals and people from low socioeconomic strata.^{36,37}

In addition to the direct effect of SARS-CoV-2 infection on mental health, the cumulative burden of chronic stress (allostatic load) has grown due to widespread pandemic-related stressors and concerns such as fear of COVID-19 infection and death, short- and long-term financial uncertainty, quarantine-associated social isolation, familial stress, and survivor's guilt.³⁸ These stressors have been further exacerbated by pre-existing stigma surrounding mental health disorders, disruptions in the health care system which resulted in diminished access to mental health services, and frequent dismissal of patients' reported mental health symptoms by the health care community, particularly early in the pandemic and for those without formal laboratory evidence of SARS-CoV-2 infection.³⁹ Moreover, *pre-existing* mental health disorders have been associated with worse clinical outcomes related to SARS-CoV-2 infection, including an increased risk of severe acute illness, hospitalization, death, PASC, and severe depression and anxiety symptoms in the setting of PASC.^{39–44}

Pathophysiology

Although the pathophysiology of SARS-CoV-2 infection is still under investigation, it is surmised that the cytokine storm created by the virus may affect mental health by initiating a central inflammatory response, reducing serotonin production which attenuates downstream monoamine neurotransmission, and increasing brain glutamate and upregulation of *N*-methyl-D-aspartate (NMDA) receptors.^{7,8} In aggregate, these changes produce damaging neuronal excitotoxicity with subsequent synaptic pruning and neuronal loss. This loss is compounded by a cytokine-mediated reduction in brain trophic factors. The cumulative effects of these disruptions may be the induction of new-onset or re-exacerbation of pre-existing neuropsychiatric conditions.^{8,9}

This role of cytokines in the context of PASC is similar to the pathophysiology of other pro-inflammatory conditions. That is, cytokines are appropriately generated in response to a peripheral infection or other process, and when they cross into the brain, they can cause normal, evolutionarily conserved, protective physiological symptoms of illness allowing the body to

cope with the infection, such as fever, lethargy, and malaise.⁴⁵ However, when activation of the peripheral immune response and pro-inflammatory cytokines is sustained beyond a normal physiological duration (e.g., in the setting of autoimmune disease, systemic infection, cancer, and/or PASC), the resulting immune signaling to the brain can lead to an exacerbation and the development of depressive symptoms in vulnerable individuals.⁴⁶ In addition, mental health symptoms specifically in the setting of PASC may also be related to other biologic factors that are caused directly by SARS-CoV-2 infection, including persistent viral reservoirs, microvascular blood clots due to endothelial dysfunction, an altered gut microbiome, and autoimmunity.^{47–50}

Patient perspective

Many patients with PASC have reported encountering stigmatization, disbelief, and assumptions that the whole of the PASC syndrome is due to their psychological state,⁵¹ despite evidence to the contrary.⁵⁰ This experience has often led to fractured relationships between health professionals and their patients, which has inherently limited the degree to which clinicians have been able to offer supportive interventions to these patients. We encourage clinicians who treat PASC to model their approach based on best practices for the treatment of patients with myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS). Specifically, PASC clinicians should take a supportive, open, non-judgmental stance; they should consider the multifaceted biological aspects to the complex syndrome and recommend medical and mental health–related treatments when a patient truly requires both types of interventions.⁵² This can be especially challenging because patients with PASC who are experiencing depression, anxiety, and/or suicidal thoughts may underreport their mental health symptoms due to concerns that their clinician will attribute all their other PASC symptoms to their mental health. This phenomenon may contribute to the trend that these patients are more likely to report receiving a significantly below-optimal level of care.⁵³ Therefore, it is helpful for clinicians to reassure patients and let them know that although not all patients with PASC experience mental health symptoms, they are one of the well-documented manifestations of PASC and have the potential to be severe.⁵³

ASSESSMENT OF MENTAL HEALTH SYMPTOMS IN PERSONS WITH PASC

Intentional and effective assessment of mental health symptoms in patients with PASC can be challenging. Some patients have experienced a delay in accurate diagnosis and appropriate management because their

TABLE 1 Mental health assessment recommendations for patients with PASC.

No.	Assessment recommendation statements
1	Patients should be screened for signs and symptoms of new or worsening depression, anxiety, or PTSD using validated tools and instruments, as appropriate. Note: Clinicians should consider suicide risk screening for patients who screen positive for mental health conditions and should have an established suicide plan in place prior to screening.
2	Clinicians should conduct a comprehensive patient history including: • Current mental health symptoms, and comparison to premorbid symptoms (if any); • Current physical and cognitive health symptoms impacting mental health symptoms that warrant further evaluation or testing and potential subspecialty referral, and comparison to premorbid symptoms (if any); • Previous experiences and responses to patient-directed resolution attempts and prescribed treatments/interventions—what has been tried, what has helped, what has exacerbated physical or mental health symptoms; • Medication history: evaluate for medications that may impact symptoms, signs, or assessment parameters (e.g., medications with anti-arrhythmic, diuretic or autonomic impact); and • Functional history to include activities of daily living, instrumental activities of daily living, school, work and avocational activities (i.e. hobbies), and quality of life.
3	Clinicians should consider a review of factors contributing to symptom presentation or exacerbation. This review should include both symptom triggers (e.g., external stressors, menstrual cycle, alcohol and drug use) in addition to psychosocial factors (e.g., financial, housing, un/employment, safety, social isolation and/or other major concerns of living) and availability of a support system and resources to provide emotional, logistical, and other support.

PASC-related symptoms have been dismissed by health care professionals as purely psychosomatic manifestations.⁶ In light of this context, we suggest that clinicians introduce the topic of mental health by validating that mental health conditions are not, in and of themselves, the cause of PASC.⁵⁰ However, mental health symptoms can be worsened by, can mimic, and can worsen, coexisting PASC symptoms, which is why assessment of mental health symptoms is essential.^{2,54} For example, generalized fatigue and post-exertional malaise as symptoms of depression can both mimic and exacerbate one another, and they can result in similar impairments to a patient's function.⁵⁵ Although beyond the scope of this statement, the PASC Collaborative developed and published guidance on cognitive and physical fatigue.^{20,22} Clinicians should take great care to identify the underlying driver(s) of functional impairment to guide appropriate treatment.

When possible, a thorough assessment of a patient's mental health–related symptoms can also be

facilitated by collateral information from one or more close friends or family members. Collateral history may especially be helpful when a patient has impaired memory, poor awareness regarding how they are functioning and/or how long their level of function has been impaired, or otherwise would have difficulty independently reporting symptoms (e.g., for sleep-related symptoms such as snoring, dozing off during the daytime, frequent nighttime awakenings, etc.).^{22,26} Loved ones can contribute important information regarding the time course and severity of observed changes to the patient's mood, coping abilities, energy level, appetite, sleep, daily function, and participation in usual activities (Table 1).

Screening

Although validated questionnaires can be helpful for efficient evaluation, if a clinician implements one or more mental health screening questionnaires, it is important for the clinician to understand whether the questionnaire assesses somatic symptoms. For instance, the Patient Health Questionnaire (PHQ-9) assesses somatic symptoms such as sleep dysfunction, fatigue, difficulty concentrating, and appetite because they can be manifestations of depressive disorders.⁵⁶ Nevertheless, these symptoms can also be present due to other PASC-related conditions such as autonomic dysfunction,^{20–26} so the PHQ-9 can overpredict the presence or severity of depressive symptoms in patients with multiple manifestations of PASC.⁵⁷ In contrast, the Patient-Reported Outcomes Measurement Information System (PROMIS) Depression scale does not inquire about somatic symptoms to reduce confounding in patients who have comorbidities (such as PASC) that affect physical health.⁵⁸ Other measures that may be used include the PHQ-2 (for depression screening), Generalized Anxiety Disorder 7 (GAD-7) (for anxiety screening), and Hospital Anxiety and Depression Scale (HADS) (for depression and anxiety screening). Regardless of the validated tool(s) used, a follow-up interview is important to clarify whether the reported mental health symptoms present across settings versus only manifest in select settings (e.g., school/work, home, social contexts), which may guide diagnosis and treatment plans.

Common screening tools for PTSD include the PCL-5 (PTSD Checklist for DSM-5) and IES (Impact of Events Scale). However, confirmation of the diagnosis may require a referral to a specialist for more extensive testing. Suspicion for PTSD should especially be raised in patients who are survivors of critical illness, experienced care in an intensive care unit (ICU),⁵⁹ had a confirmed SARS-CoV-2 infection, are frontline health care workers, and/or were quarantined. PTSD can manifest as flashback-like dissociative reactions; efforts to avoid

trauma-related thoughts, feelings, places, and/or people; persistent negative cognition and mood; and symptoms of hyperarousal, such as anxiety, sleep difficulties, and irritability.^{60–63}

The context of delivering a mental health screening tool can affect patients' responses. Although screening questionnaires are often given to patients prior to a clinical encounter and before any clinical context is provided, some patients report answering mental health-related questions more thoughtfully and honestly when: (1) a clinical team member describes the relevance and purpose of the screening tool as it is introduced to them, and (2) the clinician acknowledges the patient's responses on the tool and explains how the results impact treatment recommendations.¹¹

Of note, some clinicians express discomfort in assessing suicide ideation and depressed mood due to a lack of familiarity of immediate next steps if a patient expresses active suicide ideation. The American Psychological Association provides continuing education on assessing and intervening with patients at risk of suicide (<https://www.apa.org/monitor/2022/06/continuing-education-intervene-suicide>), and Columbia University, through the Lighthouse Project, provides training and disseminates the Columbia Protocol—also known as the Columbia-Suicide Severity Rating Scale (C-SSRS) at <https://cssrs.columbia.edu/>. Clinicians will need to establish what their plan will be for patients who endorse suicide ideation. A plan may include immediate consultation with an in-clinic mental health specialist such as a social worker or psychologist in an interdisciplinary setting, or contacting a crisis worker, suicide hotline (988 or National Alliance on Mental Illness [NAMI] hotline), or mental health crisis center for more detailed inquiry with the patient and necessary treatment or intervention to maintain safety and care of the patient. If a patient expresses active suicidal ideation, they should not be left unsupervised. It is important to assure the patient of the importance of acknowledging their level of distress and providing immediate care, which may also include transfer or referral to a hospital emergency department and/or emergency psychiatric hospital. Urgent care of suicide ideation is equally as important as other physical health crisis situations and should be managed with equal immediacy.

Unique populations

When selecting appropriate screening tools, administration approaches, and collateral history informants, it is essential to consider a patient's chronological/mental age, educational history, and social environments. For example, in patients with a low reading level (in the language in which the screening tool is being administered), screening tools may need to be read aloud rather than

self-administered. In these cases, patients may be more sensitive to the perceived social acceptability of their reported symptoms, which may introduce bias into the results of the screening tool.

Clinicians who manage patients with neurodevelopmental disorders should take an especially judicious approach to selecting assessment measurements and collateral history informants. Assessment of people with neurodevelopmental and/or sensorimotor disorders may require proxy screening inventories that are focused largely on observed behaviors rather than subjective experiences, and when possible, multiple informants should be used to characterize the patient's clinical course (e.g., symptom onset, functional changes). Nevertheless, to the extent possible based on the patient's comprehension and language abilities, both self and proxy (e.g., caregiver, teacher, etc.) assessments should be incorporated into the diagnostic evaluation. Literature regarding people with intellectual disabilities has suggested mixed agreement in self- and caregiver-report across health-related factors, with better correlation for reported subjective health than for subjective distress, physical activity engagement, and social support.⁶⁴

It is also important to consider identity factors such as race, ethnicity and sexual orientation and gender identity, and clinicians should make a conscious effort to understand their own related implicit (unconscious) biases. Clinicians must also be aware that cultural differences exist in how mental health symptoms are expressed and/or described such that patients from certain cultures may be particularly hesitant to endorse mental health symptoms due to a concern for social acceptability.⁶⁵

Comprehensive history

As appropriate, diagnostic tests should be reviewed and/or obtained to evaluate for medical conditions that can masquerade as symptoms of a mental health disorder. For example, a patient's tachycardia and palpitations can be attributed mistakenly to anxiety when they are manifestations of cardiovascular and/or autonomic dysfunction.^{23,24} A higher risk of having a medical condition that presents as, or contributes to, mental health symptoms has been reported in older adults, people with pre-existing medical conditions, people with a prior psychiatric history, and people from lower socioeconomic strata.⁶⁶

There is considerable overlap in the symptoms of neurologic and psychiatric disorders that commonly manifest in PASC. Commonly overlapping symptoms include pain, disordered sleep, fatigue, apathy, cognitive changes, and mood changes. Major categories of neurologic disease that can mimic psychiatric

conditions and contribute to mental health symptoms include neurodegenerative disorders (e.g., Parkinson disease, Huntington's disease, Alzheimer's disease), vascular disorders, inflammatory/infectious/autoimmune disorders, trauma, metabolic/endocrine disturbances, nutritional deficiencies, structural disorders, and primary sleep disorders. Post-intensive care syndrome (PICS) can also manifest as neuropsychiatric symptoms and can coexist with PASC, although movement disorders are more common in PICS than in PASC.⁶⁷ Factors that may suggest an underlying (or coexisting) neurologic disorder as opposed to an isolated psychiatric disorder include: features atypical for the psychiatric disorder, an unusual age of symptom onset, paroxysmal symptom onset, normal functioning prior to symptom onset, recent substance abuse/intoxication leading to encephalopathy, prescription medicine overuse, and/or symptoms that are resistant or respond unusually to conventional treatment.⁶⁶ The DSM-5 requires that formal diagnosis of a mental health disorder considers a careful clinical history to assess for these factors.

Because emotional wellness is bidirectionally related to a person's ability to engage in daily tasks, it is also important to consider the effects of mental health symptoms on a person's function, and vice versa. A functional history should explore the patient's baseline and current ability to engage in daily self-care activities and instrumental activities of daily living (e.g., bathing, dressing, grooming, ambulating with or without use of adaptive equipment, cooking, cleaning, shopping, attending to household chores, and paying for needs/bills). PASC-related physical and mental health symptoms can also limit a patient's ability to attend and fully participate in work, school, social, and family life activities, and these limitations can, in turn, impact mood, adaptive coping, and satisfaction with life.⁶⁸ Decreased engagement in these activities can result in added stress, financial strain, and a feeling of isolation, and patients may feel pressured to engage in these activities even if participation worsens their other PASC-related symptoms such as post-exertional malaise. Understanding this patient-specific interplay is an important prerequisite to developing a patient-centered treatment plan that addresses interrelated physical impairments, mood symptoms, and coping skills, regardless of whether symptoms were premorbid versus new-onset with PASC.

Contributing factors

Clinicians should also inquire about potential triggers that exacerbate a patient's mental health symptoms. Reported potential symptom triggers include, but are not limited to, certain foods, menstrual cycles,⁶⁹ pre-

existing medical conditions,⁷⁰ prescribed and over-the-counter medications, alcohol and drug use or misuse,⁷¹ physical or social stress, and psychosocial factors such as being uninsured or underinsured, being unemployed or underemployed, being socially isolated, having limited social support (e.g., people to assist with transportation, translation, or other needs), and having concerns regarding personal safety, food insecurity, and stable housing.^{72–74} These psychosocial stressors are more prevalent in people with PASC.^{72,73} Furthermore, mental health assessment should also consider the patient-specific impact of the COVID-19 pandemic itself, such as: (1) social stress from prolonged social distancing, family illness, and/or disruptions in school, childcare, and/or work, or (2) traumatic stress due to the frequent media coverage of life-threatening illness and/or direct exposure to other people who had severe SARS-CoV-2 infection.⁷⁵

TREATMENT OPTIONS FOR MENTAL HEALTH SYMPTOMS IN PERSONS WITH PASC

The clinical course for mental health symptoms in the setting of PASC varies widely from patient to patient. Symptoms may be more severe and prolonged in patients who had a pre-existing mental health condition and/or a more severe acute course of COVID-19.^{76,77} Available evidence suggests that the general principles of treating mental health symptoms in the setting of PASC should generally mirror the treatment principles for those who do not have PASC (see Table 2).^{78,79} Patients' symptom constellations, pre-existing and/or PASC-related comorbidities, degree of functional impairment, and the extent to which their quality of life is impacted should guide the treatment approach.

It is important to note that the treatment recommendations provided in this guidance statement should not preclude clinical judgment and must be applied in the context of each specific patient, with adjustments for patient preferences, comorbidities, and other factors. In addition, the PASC Collaborative recognizes that patients with PASC typically present with a cluster of symptoms that cross multiple body systems, such as fatigue and dysautonomia, which might limit their ability to fully participate in some of these assessment and treatment recommendations. These co-existing conditions must always be considered when suggesting a test or treatment such as physical rehabilitation. As with any treatment plan, clinicians treating patients with PASC are encouraged to discuss the unknowns of PASC treatments and prognosis, as well as the benefits and risks of any treatment approach.

TABLE 2 Treatment recommendations for patients with mental health symptoms in PASC.

No.	Treatment recommendation statements
1	For patients with symptoms that are impacting everyday life/function and are outside the scope of practice of the treating clinician, consider referral to an appropriate specialist (examples include: social work, therapist, counselor, neuropsychologist, clinical psychologist, and/or psychiatrist) with formal expertise in psychological/psychiatric assessment and treatment. Note: Clinicians should discuss with the patient that mental health symptoms can be worsened by (and can worsen) other coexisting PASC symptoms.
2	In collaboration with appropriate specialists and/or primary care, consider initiation of pharmacologic and non-pharmacologic interventions.
3	In collaboration with appropriate specialist(s) and/or primary care, treat underlying medical conditions, such as pain, insomnia/sleep disorders (including poor sleep hygiene), and neurologic issues, which may be contributing to mental health symptoms.
4	In collaboration with prescribing or primary care clinician consider reduction of polypharmacy and medications that may worsen mental health symptoms.
5	Based on overall treatment plan, consider a follow-up plan to ensure symptoms are responding to treatment and intervention is progressing.

Referral

If a patient screens positive for a psychiatric disorder using one of the aforementioned screening tools or in clinical interview, clinicians with adequate training and comfort with psychopharmacologic management may provide in-office care as a means of offering immediate intervention. However, referrals to a mental health specialist (e.g., psychiatrist, psychologist, neuropsychologist, social worker, mental health counselor, etc.) for further evaluation and management should be considered for patients who: (1) score in the moderate to severe level on formal depression and anxiety screening tools, (2) endorse non-emergent but more significant psychiatric symptoms (e.g., psychosis, passive suicidal ideation and/or intent without means, active addiction to substances, significant post-traumatic stress symptoms, comorbid psychiatric conditions such as anxiety disorder and PTSD, etc.), (3) are reporting major functional impairment and/or distress related to their mental health symptoms, and (4) are not demonstrating improvement over time, despite initial treatment within the clinician's comfort level.

When possible, consider referral to mental health specialists who have experience treating patients with PASC; however, access may be limited in some regions.¹⁰ In general, referral to an appropriate specialist that has earlier access is favorable to referring to a clinician who has PASC expertise but less timely access. Given the scarcity of psychiatric resources and

other challenges that may arise in connecting patients to appropriate mental health practitioners, clinicians are encouraged to explore options available to them within their network via institutional/organizational social work resources, local community clinics, patients' insurance provider directories, and national online directories of mental health specialists (e.g., [PsychologyToday.com](https://www.psychologytoday.com)). Many physicians are already connected with referral sources for mental health treatment not related to PASC and are encouraged to reach out to their usual contacts and local/state psychological and social work professional associations to see if there are mental health professionals who are developing specialty in PASC-related mental health treatment.

Because mental health symptoms may coexist with other PASC-related symptoms (e.g., insomnia, anergia/fatigue, cognitive disturbance, pain), referrals to mental health practitioners should be made in conjunction with referrals to other medical specialists. For example, a clinician might refer a patient with severe fatigue and insomnia to both a sleep specialist and a mental health professional. Likewise, a patient with concurrent mood and cognitive symptoms in PASC may benefit from referral for a comprehensive neuropsychological assessment to inform the differential diagnosis by elucidating potential psychological contributions to persisting symptoms.

The role of telemedicine to provide mental health support also warrants consideration and has gained popularity during the COVID-19 pandemic. During the pandemic, remote support was provided to patients via a combination of telephone/video calls, mobile applications, and internet-based programs.⁸⁰ In this way, mental health-related telemedicine may be feasible and appropriate for the support of patients, family members, and health care professionals. It also offers advantages over in-person treatment for patients with PASC who have coexisting fatigue, post-exertional malaise, disability-related conditions, and/or transportation barriers.⁸¹ Nevertheless, equity concerns remain for patients who are already often underserved, especially for patients who do not have reliable internet access and/or do not know how or do not prefer to engage with technology in this manner.

Pharmacologic treatment

For patients interested in pharmacologic therapy, clinicians may prescribe appropriate medications to address specific mental health-related symptoms. Medication selection should be guided by: (1) patient preference through a shared decision-making approach, (2) consideration of medical comorbidities that would create contraindications to certain medications, and (3) caution using medications with known side effects that may exacerbate coexisting PASC

symptoms such as fatigue and neurocognitive dysfunction. Pharmacologic treatment of psychiatric conditions in PASC is largely guided by traditional approaches to psychiatric conditions, with the addition of several special considerations for this population. Table 3 provides a snapshot of available pharmacologic options used in depression, anxiety, and PTSD in the United States. It should be noted that specific recommendations on therapeutic dosing and titration are limited with regard to what is effective in patients with PASC.

Antidepressants are the mainstay of treatment of isolated depressive disorders, whereas selective serotonin reuptake inhibitors (SSRIs) and serotonin and norepinephrine reuptake inhibitors (SNRIs) are used for anxiety disorders as well. Limited studies have examined pharmacologic treatment of depression after a COVID-19 diagnosis and have shown evidence for SSRIs including sertraline, citalopram, escitalopram, paroxetine, fluvoxamine, and fluoxetine.^{97,98} Medications that have shown evidence of effectiveness for treating depressive symptoms in ME/CFS, which is a condition with several biological and symptomatic similarities to PASC, include escitalopram, nortriptyline, amitriptyline, maprotiline, doxepin, bupropion, and moclobemide.⁹⁹ Several important considerations may guide medication choices for patients with symptoms of anxiety, depression, and other coexisting PASC symptoms. For example, for patients with:

- Dysautonomia—SNRIs and tricyclic antidepressants (TCAs) can increase cardiac sympathetic control and decrease vagal activity,¹⁰⁰ and SSRIs can cause peripheral vasoconstriction and decrease venous pooling.^{23,101}
- Mast cell activation symptoms—SSRIs have been shown to inhibit mast cell activation and proliferation.¹⁰²
- Micro-clotting in the setting of SARS-CoV-2 infection and/or PASC—recent evidence suggests that patients may benefit from SSRIs due to their effects on platelets, especially those with strong sigma-1 receptor agonist activity such as fluvoxamine.¹⁰³
- Cognitive dysfunction—bupropion may be an optimal choice due to evidence that it improves cognitive dysfunction associated with depression.^{22,104}
- Anxiety disorders that require short-acting, as-needed medications for acute symptoms—benzodiazepines typically perform this function well. However, benzodiazepines should be avoided in patients who have coexisting cognitive dysfunction and/or PTSD symptoms because they impair anterograde memory¹⁰⁵ and are associated with worse outcomes in patients with PTSD.¹⁰⁶ Alternatively, gabapentin has shown efficacy for neuropathic pain and can be used as an adjunct for anxiety, although it is also associated with cognitive symptoms.¹⁰⁷
- Fatigue related to ME/CFS—low-dose aripiprazole has shown some preliminary evidence for benefit.^{20,108}

TABLE 3 Pharmacologic agents used in depression, anxiety, and PTSD.^{82–85}

Agent	Uses	Therapeutic Dose Ranges (mg/day)	Side and adverse effects
Selective serotonin reuptake inhibitors (SSRIs)			
Citalopram	- GAD - Depression	20–40	Gastrointestinal effects (nausea, vomiting, and changes in bowel habits), sedation, sexual dysfunction, increased risk of bleeding ^{85,86}
Escitalopram	- GAD - Depression	10–20	
Fluoxetine	- GAD - PTSD - Depression	20–80	
Fluvoxamine	Depression	50–300	
Paroxetine and paroxetine ER	- Chronic PTSD - GAD - Depression	20–60	
Sertraline	- PTSD - GAD - Depression	50–200	
Selective serotonin noradrenaline reuptake inhibitors (SNRI)			
Desvenlafaxine	Depression	50	Gastrointestinal effects (nausea, vomiting, constipation), anorexia/weight loss, anxiety, dizziness, hypotension, insomnia, sexual dysfunction
Levomilnacipran	Depression	40–120	Gastrointestinal effects (nausea, vomiting, constipation), anorexia/weight loss, palpitations, hypotension, sexual dysfunction
Duloxetine	- GAD - Depression	30–60	Similar adverse effects as SSRIs. In addition, dry mouth, blurry vision, increased sweating, urinary retention, constipation, and dose-dependent increases in blood pressure (more so with venlafaxine) ⁸⁶
Venlafaxine, extended release	- PTSD - GAD - Depression	75–225	
α-Adrenergic antagonists			
Prazosin	PTSD (trauma-related nightmares and sleep disruption)	3–15 mg/night	Serious adverse effects: First dose hypotension, orthostatic hypotension, syncope, intraoperative floppy iris syndrome, priapism Common adverse effects (5% to 10% of patients): dizziness, headache, drowsiness, lack of energy, weakness, palpitations ⁸⁷
Second-generation (atypical) antipsychotic medications			
Quetiapine	- Depression - Anxiety - PTSD	100 ng/mL to 1000 ng/mL	Tachycardia, dyspnea, cough, pharyngitis, rhinitis and nasal congestion, dry mouth, constipation, dyspepsia, abdominal pain, leukopenia, neutropenia, lethargy, hyperlipidemia, hyperglycemia, peripheral edema, sedation, weight gain, tardive dyskinesia ⁸⁸
Gabapentinoid			
Pregabalin	GAD	150–600	Drowsiness, dizziness, vertigo, weight gain ⁸⁹
Other anxiolytics and benzodiazepines			
Buspirone	- Anxiety, GAD - Depression	15–60	Dry mouth, insomnia, increased anxiety, gastrointestinal upset, sweating, hypertension ⁹⁰
Alprazolam	Anxiety, GAD	1–4	Drowsiness, tiredness, dizziness, sleep problems (insomnia), memory problems, poor balance or coordination, slurred speech, trouble concentrating, irritability, diarrhea, constipation, increased sweating, headache, nausea, vomiting, upset stomach, blurred vision, appetite or weight changes, swelling of hands or feet, muscle weakness, dry mouth, stuffy nose, loss of interest in sex, worsening depression, hypomania, decreased mental alertness ⁹¹

(Continues)

TABLE 3 (Continued)

Agent	Uses	Therapeutic Dose Ranges (mg/day)	Side and adverse effects
Lorazepam	• Anxiety, GAD	2–10	Sedation, dizziness, asthenia, ataxia, respiratory depression, hypotension, fatigue, amnesia, confusion, disinhibition, irritability, libido changes, menstrual irregularities, diplopia, dysarthria, appetite changes, constipation, incontinence, urinary retention, dystonia, AST and ALT elevation, In rare instances, lorazepam can cause acute liver injury (cholestatic pattern). ⁹²
Chlordiazepoxide	• Anxiety, GAD	15–100 (multiple doses)	Fatigue, sedation, depression, dizziness, ataxia, slurred speech, weakness, confusion, forgetfulness, nervousness, hyperexcitability, and pain at the injection site ⁹³
Oxazepam	• Anxiety, GAD	30–60	Sedation, fatigue, depression, confusion, memory impairment, dizziness, ataxia, slurred speech, hyperexcitability, nervousness, and weakness. Less frequently encountered adverse effects may involve hypotension, hallucinations, mania, dry mouth, hypersalivation, edema, leukopenia, decreased libido, incontinence, rash, menstrual irregularities, jaundice, and diplopia. ⁹⁴
Serotonin-modulating antidepressants and others			
Bupropion	• Depression	150–450	Tachycardia, insomnia, agitation, dizziness, tremor, constipation, dry mouth, rhinitis, pharyngitis, headache, diaphoresis, weight loss, nausea, blurred vision ⁹⁵
Mirtazapine	• PTSD • GAD • Depression	15–45	Weight gain, sedation (appears to be less marked at higher treatment doses) ⁹⁶
Vilazodone	• Depression	20–40	Nausea, vomiting, diarrhea, insomnia
Vortioxetine	• Depression	5–20	Nausea, vomiting, diarrhea, constipation, flatulence, dry mouth, dizziness, unusual dreams, sexual dysfunction in males and females

Abbreviations: GAD generalized anxiety disorder; PTSD post-traumatic stress disorder.

- PASC-related mania—manic symptoms have been shown to respond well to atypical and typical antipsychotics, mood stabilizers, and benzodiazepines.¹⁰⁹
- Psychosis—generally treatment with first- or second-generation antipsychotics. Second-generation antipsychotics such as aripiprazole, olanzapine, and risperidone have immunomodulatory¹¹⁰ effects that may be beneficial in the setting of SARS-CoV-2 infection and PASC.

Non-pharmacologic treatment

The foundation of non-pharmacologic management of PASC should arise from the therapeutic alliance between the clinician and the patient. A positive, collaborative relationship is created when the patient feels both safe and empowered. Perhaps one of the most important actions a clinician can take when establishing this relationship is the validation of the patient's concerns and reassurance that they are being taken seriously. It is helpful to give patients a reasonable degree

of control over treatment decisions when appropriate, which allows them to have agency in their recovery process. It is also necessary to communicate with the patient that the goal of mental health treatment is to minimize the degree to which mental health symptoms are impacting the person's illness experience.

Efficacy for specific non-pharmacologic treatments for mental health symptoms in the setting of PASC has not been well-described in the current medical literature. However, the biopsychosocial intervention approach, which includes behavioral health treatments, has been found to be effective in reducing mood symptoms and alleviating physical symptoms for conditions with similar features as PASC (e.g., chronic fatigue and pain). Useful approaches for individual psychotherapy commonly include a combination of supportive and cognitive behavioral therapy (CBT) modalities. Supportive psychotherapies have evidence for use in medically ill populations to improve patient functioning and decrease the burden of psychological distress.¹¹¹ A CBT approach (including acceptance and mindfulness-based treatments) may be beneficial for patients who

report, or have evidence of, premorbid somatic preoccupation, perceived stress and loneliness, clinical depression and anxiety, and a high degree of worry about COVID-19.¹¹² These features are disproportionately prevalent, although not ubiquitously present, in people who develop PASC.^{113,114} In patients with PTSD, exposure therapy should be considered and performed with caution because proper autonomic nervous system function is necessary to safely participate in this modality, and dysautonomia is prevalent in this population.^{115,116} The Substance Abuse and Mental Health Services Administration (SAMHSA) Advisory for PASC lists other condition-specific, non-pharmacologic therapeutic strategies and hyperlinks to resources (Appendix B: Assessment Tools and Therapeutic Strategies by Functional Domain).

Group psychotherapy may also be beneficial for patients with PASC because it can provide social connection, validation, acceptance, awareness of additional resources, and hope.¹¹⁷ In addition, group therapy is a good setting for efficiently teaching adaptive coping skills that are helpful to improve patients' mood, self-efficacy, self-care abilities, and quality of life. Of note, both before and during the COVID-19 pandemic, people who identify as a member of a marginalized group (and especially with racial and ethnic minority groups) have experienced disparities in access to mental health and rehabilitative care, quality of the care delivered, and clinical outcomes achieved.^{118–120} Every effort should be made to close gaps in these health disparities and ensure optimal care for this population. Public health initiatives should provide culturally responsive support to these minoritized groups. Appendix C of the SAMHSA Advisory for PASC lists support groups, patient advocacy groups, and other resources for people with PASC and their family members and caregivers. Some of these resources are designed specifically for people who belong to specific underrepresented groups.¹²¹

Of note, some patients may be interested in self-help resources, perhaps because of financial or scheduling-related barriers, because they do not yet feel ready to discuss their mental health symptoms with a clinician, and/or because they would like to supplement the professional mental health care they are seeking. Clinicians can provide these patients with free information from nonprofit organizations such as the NAMI (www.nami.org/Support-Education/Mental-Health-Education) and Mental Health America (www.MHANational.org/Self-Help-Tools). Several free and subscription-based mental health-related mobile applications (apps) also offer stress-management techniques (e.g., meditation, mindfulness-based stress reduction, guided imagery, progressive muscle relaxation, deep breathing) and CBT-related features. Clinicians should advise patients who are considering using a mental health app to consider the app's cost, privacy

and security policies, and usability. Clinicians may want to consider using the app as a trial themselves prior to offering individual suggestions to their patients, keeping in mind their own biases and cost considerations, and the use of these apps should be encouraged in addition to, rather than in place of, psychotherapeutic treatment with a trained mental health clinician. The U.S. Department of Veterans Affairs maintains an online health library with free apps to assist with stress and mindfulness, as well as resources on COVID, depression, and PTSD. These apps are easy to download, navigate and use (<https://www.veteranshealthlibrary.va.gov/HealthyLiving/Stress/>).

Treatment of underlying medical conditions

Because mood and anxiety disorders may be secondary to, exacerbated by, and/or a contributing factor to other medical conditions,²² it is important to address coexisting medical conditions to reduce a patient's overall mental health symptom burden. Although the identification and treatment of coexisting medical conditions is straightforward in some cases, in other cases it is not. The aggregate clinical presentation of PASC has features and symptom presentations that are similar to other post-infectious syndromes (e.g., after infection with Epstein–Barr Virus [EBV], *Borrelia* bacteria [Lyme disease], or Zika virus) and clinical syndromes without a definitive infectious precursor (e.g., ME/CFS, fibromyalgia, PICS, and postural orthostatic tachycardia syndrome [POTS]). It is hypothesized that the common symptom constellation including fatigue, disordered sleep, chronic pain, headaches, cognitive symptoms, and/or mood disorders may be a manifestation of a final common pathway across all these conditions that results in a phenotype of post-infectious, immunologic, and neuronal dysregulation. Regardless of the underlying medical condition that is contributing to these symptoms, management can include supportive care (e.g., pacing, adaptive strategies) and lifestyle-based interventions (e.g., optimizing sleep hygiene, hydration, and sodium/nutritional status).

Polypharmacy reduction

Some of the most commonly reported PASC symptoms, including those that can be manifestations of a mental health condition (e.g., fatigue, cognitive dysfunction), can also be exacerbated iatrogenically due to polypharmacy and/or medication side effects.^{42,122,123} In many cases, medication de-prescribing (including dose reduction and/or discontinuation) is a preferred solution. Substantial precedents for adverse drug events exist in other vulnerable populations such as

older adults and patients with dementia and/or other chronic illnesses. As a result, deprescribing guidelines (such as the BEERS criteria list) have been developed previously and can provide a framework for a similar deprescribing approach in PASC.^{124–126} Classes of medications to consider deprescribing include sedatives/hypnotics (e.g., barbiturates, benzodiazepines), anticholinergics, antihistamines, antidepressants or antipsychotics with anticholinergic activity, pain medications with potential sedating effects, and skeletal muscle relaxants (which have sedating and anticholinergic effects). Many medications used to treat mental health symptoms also have such properties; therefore, this issue is of particular relevance in the setting of PASC.

Another polypharmacy consideration in this population is that many medications that are used to treat mental health symptoms can impact underlying medical issues. Examples include medications that have antihypertensive effects (e.g., atypical antipsychotics, alpha-2 agonists, beta blockers), blood pressure raising effects (e.g., venlafaxine, psychostimulants), and effects on heart rate (e.g., psychostimulants, beta blockers). Inadvertently lowering blood pressure or heart rate in the setting of PASC may exacerbate a patient's underlying fatigue and/or dysautonomia and could cause syncopal events. Conversely, people with PASC-related dysautonomia may also have an increased risk of tachycardia associated with psychostimulant use.^{115,116} An additional risk of stimulant use in this population is that patients may experience worsened post-exertional malaise if the stimulant causes them to temporarily feel more alert and capable of carrying out daily activities but then they unintentionally work past their energy threshold. There may also be considerations for deprescribing. Resources on deprescribing can be found at <https://deprescribing.org/>.

Follow-up

Follow-up plans can be guided by patients' symptom severity, symptom trajectory, response to treatment, and social factors such as their health literacy, available family and community support, and ability to advocate for themselves. Patients with more significant and/or complex mental health symptoms and related impairment should receive earlier and more frequent follow-up evaluations. However, for most patients who present with less severe symptoms, reassessment of mental health symptoms can occur in the context of routine medical follow-up.

SUMMARY AND FUTURE DIRECTIONS

In summary, new mental health symptoms can be caused by PASC, worsening of pre-existing mental

health symptoms can be caused by PASC, and mental health conditions are not, in and of themselves, the overall cause of PASC. Although it may sometimes be unclear whether PASC or mental health symptoms came first, it remains vital to prioritize assessment and management of existing symptoms. Initiating the clinician/patient encounter with a discussion that validates the patient's experience and describes this management approach is an essential step to effectively and accurately screen for and assess mental health symptoms. This discussion can also raise awareness, reduce stigma, and be the first step to the patient seeking dedicated mental health treatment. Management of mental health symptoms in the setting of PASC must also consider the patient's individualized cluster of symptoms because other PASC-related symptoms such as fatigue and dysautonomia might limit someone's ability to fully participate in treatment recommendations that would otherwise be considered first line.

Although evidence is emerging, there continues to be a knowledge gap regarding how PASC-related mental health symptoms affect patients from various sociodemographic subgroups (e.g., based on biological sex, gender identity, racial or ethnic group, socioeconomic background, health insurance status, or occupation).¹²⁷ More research regarding PASC-related manifestations of mental health disorders is also necessary for special populations including children and people with complex medical considerations such as chronic immunosuppressive therapy, organ transplantation, and hemodialysis.¹²⁷ For PASC-related mental health interventions to reach the largest population of patients with the greatest need, we advocate that social factors that contribute to patients' stressors and affect their access to care also be addressed.

AAPM&R HEALTH EQUITY STATEMENT

The American Academy of Physical Medicine and Rehabilitation (AAPM&R) recognizes the need to support equitable access to rehabilitation care for individuals with Post-Acute Sequelae of SARS CoV-2 infection (PASC). The AAPM&R states that equitable access to care includes: (1) timely and local patient access to multidisciplinary care; (2) addressing inequities in the United States health system that result in diminished access to sustained quality care because of structural racism or socioeconomic factors; and (3) strengthened safety-net care, including disability evaluation and benefits.¹²⁸

Each of the AAPM&R's PASC guidance statements were produced by a diverse and multidisciplinary team of subject matter experts with patient input. Although an in-depth discussion of health equity issues is beyond the scope of the PASC guidance statements, each one highlights health equity concerns and refers readers to

other publications and resources. The term “health equity” has many different definitions, and they generally focus on ensuring that every person is able to achieve the highest level of health and function. For example, the Centers for Disease Control and Prevention (CDC) defines health equity as the opportunity for people to fulfill their full health potential and states that people should not be disadvantaged from achieving their potential because of social position or other socially determined circumstances.¹²⁹ The Centers for Medicare & Medicaid Services (CMS) uses the definition established in Executive Order 13985, issued on January 25, 2021, which states that equity is “the consistent and systematic fair, just, and impartial treatment of all individuals, including individuals who belong to underserved communities who have been denied such treatment, such as Black, Latino, and Indigenous and Native American persons, Asian Americans and Pacific Islanders and other persons of color; members of religious minorities; lesbian, gay, bisexual, transgender, and queer (LGBTQ+) persons; persons with disabilities; persons who live in rural areas; and persons otherwise adversely affected by persistent poverty or inequality.”¹³⁰ There are many root causes for health disparities, some of which fall under the categories within social determinants of health (SDOH). Examples of SDOH include but are not limited to socioeconomic status, neighborhood, availability and access to healthy food, and access to a high-quality education.

In addition to advocating for equitable access to rehabilitation care for all persons with PASC, the AAPM&R supports four “Principles of Inclusion and Engagement” which include: (1) valuing diverse group composition (a diverse group is more representative of AAPM&R’s membership and volunteers may be selected as a member of a particular community to enhance diversity of thought and experiences); (2) mutual respect (cultivating a receptive space for differing opinions and viewpoints); (3) talent and skill-based selection for leadership opportunities (ensuring that broad criteria of diversity of experience, talent and knowledge are incorporated and removing barriers to involvement that support an equitable environment); and (4) comprehensive collaboration (building community among various member constituent and bringing together different perspectives).¹³¹ Readers of the PASC guidance statements are encouraged to consider the recommendations through the lens of health equity in order to improve access to rehabilitation care for all individuals with PASC.

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ORCID

Abby L. Cheng  <https://orcid.org/0000-0002-8166-0806>

Talya K. Fleming  <https://orcid.org/0000-0002-2034-8253>

Rowena Ng  <https://orcid.org/0000-0001-7193-4300>
 Sarah Sampsel  <https://orcid.org/0000-0002-7496-8482>
 Monica Verduzco-Gutierrez  <https://orcid.org/0000-0003-0964-5908>

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